

Project Development Phase

AI Powered Video Script Generator

1. Artificial Intelligence

Artificial Intelligence (AI) enables machines to perform tasks that normally require human intelligence such as understanding language, reasoning and generating content. In our project, AI is the heart of the application. The user provides inputs such as topic, target audience, tone and video duration. These inputs are processed through the Groq API, which accesses the pre-trained LLaMA 3.3 model. The model understands the prompt and generates a structured promotional video script including a hook, body, scene suggestions and call-to-action. AI removes the need for manual script writing, improves creativity, reduces effort and provides results within seconds. This makes the application useful for students, marketers, startups and content creators.

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2. Machine Learning

Machine Learning is a branch of AI in which models learn from large amounts of data. Our project does not train a new ML model; instead it uses a pre-trained Large Language Model that has already been trained on huge datasets. This saves development time and computing resources while still providing high-quality outputs. ML concepts remain important because the LLM itself is the result of machine learning.

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3. Generative AI, Functional and Performance Testing

Generative AI creates new content rather than only analysing existing data. In our project it generates original video scripts based on user prompts. Functional testing included verifying topic input, tone selection, audience

selection, duration, API connectivity, error handling and output formatting. Performance testing checked response time, consistency, reliability and behaviour under repeated requests. These tests ensured the application generated useful scripts accurately and efficiently.

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4. Power BI

Power BI was not implemented in the current version. However, it could be integrated to analyse user activity, popular video categories, API usage, daily script generation and dashboard reports. Such insights would help improve the application and understand user behaviour.

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5. Tableau

Tableau is another business intelligence tool for interactive data visualisation. Although not used directly, it could display trends such as script generation frequency, most-used tones, audience preferences and monthly usage reports through charts and dashboards.

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6. Salesforce

Salesforce is a CRM platform. It is not part of the current implementation, but future versions could integrate Salesforce to manage registered users, customer subscriptions, feedback, marketing campaigns and business leads if the application is commercialised.

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7. User Acceptance Testing (UAT/FSD)

User Acceptance Testing ensures that the application satisfies end-user expectations. Test scenarios included generating scripts for different topics, checking output quality, validating incorrect inputs, verifying response speed and confirming that the generated script matched user requirements. Feedback suggested the application was easy to use, generated relevant scripts and significantly reduced script-writing time.

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Conclusion

The AI Powered Video Script Generator demonstrates the practical use of Artificial Intelligence and Generative AI to automate promotional script creation. The project successfully integrates Flask, Groq API and the LLaMA model to provide fast and meaningful results. While tools such as Power BI, Tableau and Salesforce were not implemented, they offer valuable opportunities for future enhancement through analytics, reporting and customer management.

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